Proving what a BerkeleyGW calculation computes

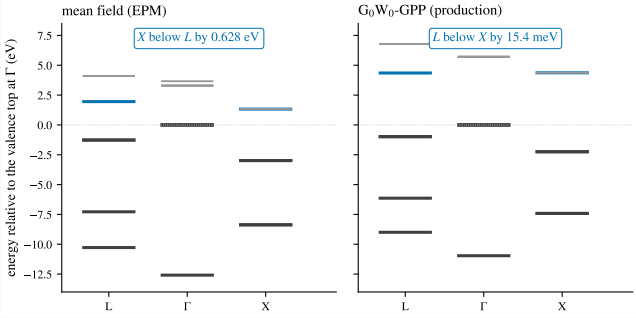
A kernel-checked certificate for a G_0W_0–GPP run on Si-EPM, matched to 5 μeV

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Formalization, certificates and analysis code written by Claude (Anthropic) under the authors' direction. Reviewed by a separate AI reviewer (Codex); not a human-maintainer audit.

The question



BerkeleyGW's own `testsuite/Si-EPM`: G_0W_0 reverses the X/L conduction ordering.

The flip is 15 meV, out of self-energies of $\sim$9 eV.

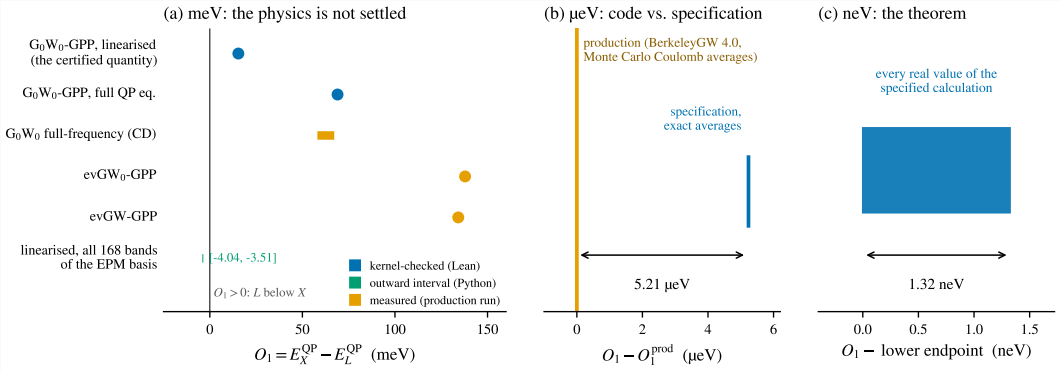
Is 15.4 meV exactly what the code's equations give for this input?

Three different questions

	question	here
1 implementation	does the code compute its specified calculation?	**proved**
2 convergence	is that calculation converged?	measured only
3 model	is G_0W_0/GPP the right physics?	not claimed

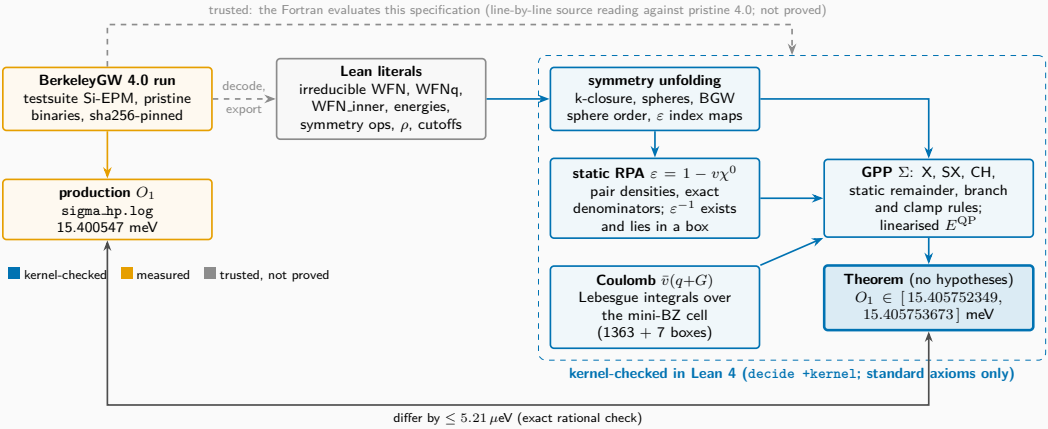
This talk is about layer 1, for one input deck.

Result



- Lean theorem, no hypotheses: $O_1 \in [15.405752349,\ 15.405753673]$ meV
- BerkeleyGW 4.0 prints: 15.400547 meV ($5.21\,\mu$eV away)

What is proved, what is trusted



The theorem

```
theorem o1_enclosure_irr_unconditional :
    O1 glI stXI stLI EIrr vbTrue oqTrue ∈
      Icc ((8880811588953599 : ℝ) / 2 ^ 59) ((8880812351936001 : ℝ) / 2 ^ 59)
```

- `O1` is a real number: exact χ^0, the true ε^{-1}, GPP Σ, linearised QP.
- Inputs: the *irreducible* wavefunctions; unfolding and Coulomb integrals are proved.
- Kernel-checked (`decide +kernel`); only the three standard axioms.

The specification reads like `mtxel_cor.f90`

BerkeleyGW 4.0, occupied GPP branch

```
if (wdiffr.gt.limittwo .and. delwr.lt.limitone) then
  sch = delw * I_eps_array(ig,my_igp)
```

Lean, `SigmaSpec.schM`

```
if (x - w) ^ 2 > GAM2 ∧ w ^ 2 < (x - w) ^ 2 * LIM1
then w * IeU I / (x - w) else 0
```

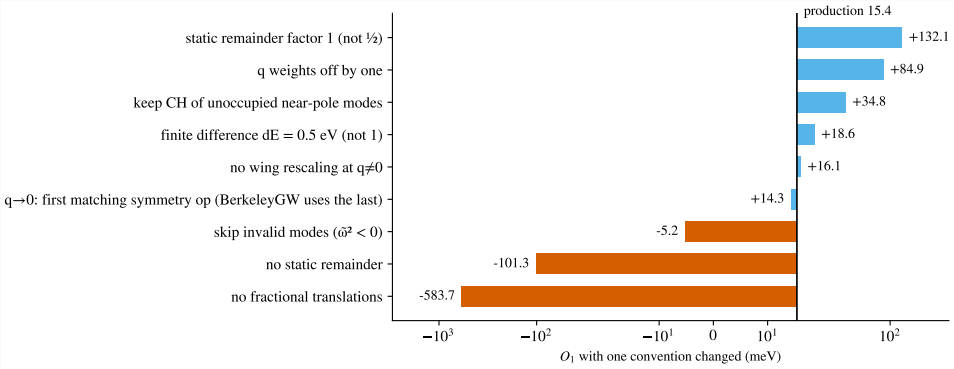
Division-free form, proved equal to the source's rule where that is defined. Every definition sits next to its Fortran in `docs/review/index.html`. That correspondence is read, not proved.

Where the 5.21 μeV comes from

- BerkeleyGW averages $8\pi/|q+G|^2$ over the mini-BZ by Monte Carlo; the proof uses the exact integrals. The averages differ by up to 1.7×10^{-3}.
- Each E^{QP} moves by 0.14–0.18 meV; in O_1 this cancels to 5.21 μeV.
- With BerkeleyGW's own Monte Carlo averages, an outward-rounded evaluator encloses every printed E^{QP}.

No implementation discrepancy on any active code path of this deck.

Documented conventions decide the sign



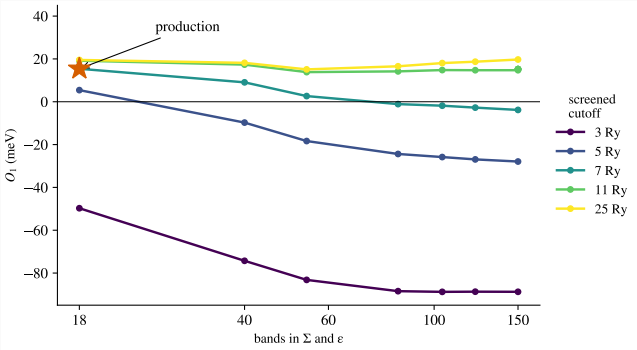
Each bar changes one convention; all 15 changes are detected at 10^{-8} eV in the Σ table. None is a bug.

The $q \to 0$ cell picks one direction

- Σ uses the *last* matching symmetry operation for each q; at $q = 0$ that fixes one direction of the anisotropic head and wings.
- Over all directions: $O_1 \in [14.34, 15.41]$ meV. Direction average: 14.89 meV.
- Exactly degenerate states split; the Γ valence SX$-$X spreads by 36 meV before the multiplet average hides it.

Outward-interval (Python) results, not kernel-checked.

Convergence: the sign comes from cancelling truncations



At 7 Ry, adding bands takes O_1 from $+15.4$ to -3.8 meV.

At 18 bands, raising the cutoff looks converged.

125 production runs. Measured; no convergence claim.

The GW hierarchy moves O_1 by tens of meV

treatment	O_1 (meV)
G_0W_0-GPP, linearised	$+15.4$
G_0W_0-GPP, full QP	$+69.0$
full frequency (CD)	$+58$ to $+67$
evGW_0-GPP	$+137.9$
evGW-GPP	$+134.2$

- $V_{xc} = 0$: QP energies sit $\sim$7 eV from mean field, deep in Σ's poles.
- Proved: exactly two QP roots per state in $[-10^4, 10^4]$ eV; the principal one is unique only because of the `sexcut` clamp.
- Plasmon-pole error at E_o: -31.6 meV.

From theory to the specification

- Proved under named hypotheses: the pole signs of Σ from time-ordered G_0, the HL-GPP algebra and SX/CH split, the static RPA structure.
- Every departure of code from theory is listed: *proved equal* or *named approximation*.
- Formalization made three edge cases explicit:
  - $x = 0$ exactly at $q = 0$ (`TOL_Zero` substitution);
  - the GPP kernel is discontinuous at $(x, \Omega^2) = (0, 0)$;
  - the one-system theorem cannot be applied at X or L.

Conditional identities; not instantiated on the certified run.

What is not claimed

Trusted, not proved

- export of the run files into Lean
- that the Fortran evaluates the specification
- the production execution

Not claimed

- correctness on other inputs
- convergence of 18 bands
- that GW/GPP or the EPM describe Si
- verification of BerkeleyGW's physics

Trusted computing base: Lean 4.19 kernel, Mathlib, elan. No external re-checker run yet.

What we would like from you

1. **Does the specification match your reading of the code?**
   Every Lean definition is next to its Fortran lines in `docs/review/index.html`.
2. **What should come next?**
   LDA V_{xc}, complex flavour, full frequency, a 2D material with truncation.
3. **Would certified reference values help the testsuite?**
   The convention study already detects changes at 10^{-8} eV.

Reproduce

```
bash scripts/bootstrap.sh
bash scripts/verify.sh
```

Build, exhaustive axiom audit,
699 scientific and 16 workflow tests;
about 5 h from a cold build.

theorems audited	34 553
certificate modules	1 376
Coulomb boxes proved	1 363 + 7
production runs	125

`github.com/MichealWangYZ/bgw-lean-certificate`

The code computes what its source says,
to 5 μeV, on this deck.

Convergence, the plasmon pole and the vertex remain physics.

References

- M. S. Hybertsen and S. G. Louie, Phys. Rev. B **34**, 5390 (1986).
- J. Deslippe *et al.*, Comput. Phys. Commun. **183**, 1269 (2012).
- B.-C. Shih *et al.*, Phys. Rev. Lett. **105**, 146401 (2010).
- The Lean 4 theorem prover and Mathlib.

Backup: evidence for each number

number	evidence	where
$O_1 \in [15.405752349, 15.405753673]$ meV	Lean	`Headline.o1_enclosure_irr_unconditional`
$O_1 > 0$	Lean	`Headline.o1_pos_irr_unconditional`
full-QP $O_1 \in [68.9856742, 68.9856744]$ meV	Lean	`Headline.o1_fullQP_irr_unconditional`
5.21 μeV	exact rationals	`tests/test_endpoint_e1.py`
$q \to 0$ range, direction average	interval	`bgw/endpoint/cert.py`
all-band $O_1 \in [-4.05, -3.51]$ meV	interval	`bgw/r28_f3`
convention study	reproducer	`data/mutations_e1.log`
convergence, FF, evGW	production	`docs/ENDPOINT.md`